# COLL #128: The Art of Lazy Programming
# Spring 2026

## Lecturer & Sponsor Information

*Name:* Charlie Cruz
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*Class Sponsor:* Dr. Vicky Yao
*Sponsor's Email:* vy@rice.edu

## Class Information

*Time:* W 6:00–7:00 PM
*Classroom:* Hum 128
*Office Hours:* W 8:00–9:00 PM

Additional individual meetings to discuss course material or any other concerns can be scheduled by contacting me via email, after class, or during office hours. Also on Discord @ SeniorMars#2916.

## Course Description

In your time as a Computer Scientist at Rice, you will learn many theoretical aspects of Computer Science and the way to apply programming to complex topics. However, another crucial step towards becoming a great Computer Scientist is efficiency as a developer. *The Art of Lazy Programming* is a one-credit course where students will be taught precisely that: students will learn vital information every programmer should know, tips on learning complex but time-saving tools, and most importantly, the answer to **"how does one master their programming environment and gain the knowledge needed to be an experienced programmer"?**

The intention of this class is to teach students various tools and topics such as grep, docker, and sshfs – that will make their life easier by introducing efficiency. While students may learn some of these tools in classes and spend dozen of hours using them, their core benefits are never explicitly taught. Yet, mastering these tools will allow students to solve large-scale problems that seem impossibly complex. Additionally, upon completion of the course, students will also develop problem-solving skills that can be applicable outside Computer Science.

## Course Objectives and Learning Outcomes

In this course, students will:

- Gain mastery over the command-line as an extension of problem-solving through efficiency.
- Efficiently search information, read codebases, and understand certain design choices.
- Develop techniques to profile, debug, and write efficient programs that apply to any language.
- Understand in-depth how to use common programming websites, tools, and services to their full advantage.
- Create their own opinions about software methodology, programming languages, and open software.

**Prerequisites**: No previous knowledge is needed, however, this class is designed to cover various different topics every week, so it would be advised to have taken one CS class **at the minimum**.

## Materials & Resources

The class website is located at lazy.rice.edu and the link to our class Github repo is: https://github.com/orgs/RTAOLP/teams/spring2026. All class notes will be posted here along with the homework for that week. Similar classes with their exercises and classnotes can be found at https://missing.csail.mit.edu/ and https://www.cs.cmu.edu/~07131/f21/. You may also find some of my past youtube videos helpful: Open Source Playlist. Don't hesitate to ask any questions if you have any.

## Grading

At the end of the course students will either get a "Satisfactory" or an "Unsatisfactory" based on:

| | |
|---|---|
| *Homework Exercises (10x)* | *55%* |
| *Final Presentation* | *30%* |
| *Attendance & Participation* | *15%* |

## Homework Assignments

Over the semester, you will be assigned 10 assignments. Each assignment will have a hands-on exercise and a reading component relating to that weeks lecture. For the former, you are expected to complete and submit your assignment to that weeks repository, and to check your solutions through the tester program. To complete the latter component, you will be asked to submit your notes/summary of that weeks reading(s) through the same method mentioned beforehand. I will not read everyones submission for this component, instead, it will mostly be graded on completion. More information will be provided in the first lecture of the semester. I will try my best to make each weeks homework combined take **1-hour max per week**. Please contact me if you do not believe I stayed true to my word. Additionally, your lowest grade will also be dropped. Finally, you are more than welcome to use man or duckduckgo to help you solve these assignments.

## Participation and Classroom Etiquette

You are expected to complete all assigned homework, come to class on time, and be prepared to participate actively. Civility in the classroom and respect for the opinions of others is very important for a learning environment. It is likely you may not agree with everything that is said or discussed in class; respectful and courteous behavior, as well as responses, are expected at all times.

## Attendance, Absences, Late Work, and Extensions

Attendance is expected at each class as lectures will be crucial for understanding and completing homework. However, I understand that people are busy, as such, you will have a free unexcused absence, and all class content can be reviewed with videos that will be posted to the class website. Extensions can be arranged as long as you email me before the deadline, and late work will be accecpted with a 10% penalty for each day it is late. However, three unexcused absences will result in an "Unsatisfactory" grade for the class.

## Final Presentation

Two weeks of this class will be dedicated to final presentations. This will be the final for this course, where each student will be asked to present a five-minute presentation about a tool, service, or website relating to our classs topic and explaining how to use it. The rubric for the final will be released near the end of the semester.

## A Typical Day In Class

It may be helpful to some to know how class is conducted. This may change, but this is the vision: First, class starts by introducing a problem in daily development. Students would then begin to discuss why such a problem exists and how to overcome that problem. As a follow-up, we would move on to the lecture on a tool that can help us solve this problem efficiently. Finally, students would turn to an opinion article relating to that week's topic and end with another class discussion.

## Unrelated Computer Science Lectures

I will give extra presentations about some CS topics I find interesting during the semester. I will give out notices to these lectures, and while these talks may not help your grade, I hope youre able to attend them. If you send me a picture of a cute platypus, Ill give you extra credit and bump any homework grade to its next grade.

## Mental Health Week Pass

If you ever feel that for one reason or another, you cant do the homework for that week or just need a breather, email me and Ill give you 100% for that week. No questions asked. **The homework is there to help you learn, not for you to suffer**. Of course, if you need someone to talk to, Im more than willing to listen.

## Rice Honor Code

In this course, all students will be held to the standards of the Rice Honor Code, a code that you pledged to honor when you matriculated at this institution. If you are unfamiliar with the details of this code and how it is administered, you should consult the Honor System Handbook at http://honor.rice.edu/honor-system-handbook/. This handbook outlines the University's expectations for the integrity of your academic work, the procedures for resolving alleged violations of those expectations, and the rights and responsibilities of students and faculty members throughout the process.

## Academic Accommodation

If you have a documented disability or other condition that may affect academic performance you should: 1) make sure this documentation is on file with the Disability Resource Center (Allen Center, Room 111 / adarice@rice.edu / x5841) to determine the accommodations you need; and 2) contact me to discuss your accommodation needs.

## Artificial Intelligence Policy

The use of AI is permitted in this course, provided that its use is properly cited and does not violate the Rice Honor Code.

## Syllabus Change Policy

This syllabus is only a guide for the course and is subject to change with advanced notice.

## Class Schedule

The schedule is tentative and subject to change.

| Lecture | Date (Wednesdays) | Topic(s) |
|---|---|---|
| 0 | January 14th | Class Intro, Unix, and Terminal Usage |
| 1 | January 21st | Shell Scripting and the Unix way |
| 2 | February 4th | Terminal Text Editors & Better Alternatives |
| 3 | February 11th | SSH, TMUX, and Servers |
| 4 | February 18th | Version Control & Advance Git |
| 5 | February 25th | Github, other Programming Websites, and Open Source |
| 6 | March 4th | Artificial Intelligence & LLMs |
| 7 | March 11th | Automation Through Programming |
| 8 | March 25th | Cryptography & Security |
| 9 | April 1st | Potpourri |
| 10 | April 8th | Final Presentations Part 1 |
| 11 | April 15th | Final Presentations Part 2 |
| 12 | April 22nd | Discussion about Software Development and its impact |